

Characteristics of liquid product from the pyrolysis of waste plastic mixture at low and high temperatures: Influence of lapse time of reaction

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Abstract

Pyrolysis of a waste plastic mixture (high-density polyethylene: low-density polyethylene: polypropylene: polystyrene = 3:2:3:1) into a liquid product was carried out in a stirred semi-batch reactor at low (350 °C) and high (400 °C) temperatures. The effect of lapse time of reaction in the reactor and also degradation temperature on the characteristics of the liquid product from pyrolysis of the mixture was investigated. Liquid products were described by cumulative amount distribution, paraffin, olefin, naphthene and aromatic (PONA) distribution and molecular weight distribution. Their characteristic was quite differed with a lapse time of reaction and also at a low and high degradation temperatures, because of the different physicochemical properties of the plastic types in the mixture. With increase of lapse time of reaction, the order for the main products in PONA components obtained at 350 °C was firstly aromatic products and then olefin products, while at 400 °C the order was firstly aromatic products, then olefin products and finally paraffin products. The experiments also showed from the molecular weight distribution of liquid PONA components that the paraffin and olefin products had a wide distribution by mainly random scission of polymer, but in the case of olefin products were produced by an end-chain scission mechanism as well as random scission mechanism, as evidenced by much more light olefin products. This phenomenon was evident at a higher degradation temperature. Also, both the light olefin and naphthene products with a molecular weight of around 120, as a main product, showed a similar trend as a function of lapse time, which had a maximum fraction at 343 min (at 350 °C) and 83 min (at 400 °C). Among PONA components, the highest concentrations of aromatic products were obtained with a molecular weight of around 100 at the fastest lapse time of reaction, regardless of degradation temperature. It was concluded that the characteristics of liquid product on the pyrolysis of plastic mixtures were strongly influenced by lapse time of reaction and degradation temperature.

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1. Introduction

Waste plastic mainly consists of polyethylene (PE), polypropylene (PP) and polystyrene (PS) with different polymer structure, which is above 70% in waste plastics (Aguado and Serrano, 1999; Ciliz et al., 2004; Williams and Williams, 1999). Thus, recycling of waste plastic is highly encouraged. With respect to the method of recycling, plastics can be practically recycled either mechanically towards a new plastic with a lower grade application, i.e., physical

recycling, or chemically to obtain chemical raw materials or fuel oil, i.e., chemical recycling. However, physical recycling, as a conventional method to treat the waste plastic mixture, is limited by technical problems, difficulties in maintaining the product quality and adequate price (Ciliz et al., 2004; Karagoz et al., 2003). Thus, physically recycled plastics should be treated by other procedures such as chemical recycling and energy recovery.

In Korea, the amount of waste plastics generated in 2002 was about 3.44 million tons, according to data from the National Institute of Environmental Research. Regarding the method of treatment or disposal of waste plastics, incineration and landfill methods were about 50 and

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30 wt%, respectively, and the other 20 wt% was through recycling. The majority of recycling methods was material recycling, although the quality of the recovered plastic was very low; only a small percentage was treated by pyrolysis and gasification (chemical recycling). However, material recycling will reach its limits, while chemical recycling will be gradually expanded.

For recycling of waste plastics, pyrolysis as a chemical recycling method is becoming a promising alternative, in order to recover fuel oil and hydrocarbon feedstock (Walendziewski, 2002; Buekens and Huang, 1998; Murata et al., 2002; Demirbas, 2004). As an advantage of this process, pyrolysis can treat all thermoplastic mixtures consisting of various types of plastics, without separation or treatment (Vasile et al., 2001). Also, it does decrease a great social resistance due to air pollution associated with incineration and soil contamination in the landfill (Buekens and Huang, 1998), and the economic resistance due to the increase of space and disposal costs. Pyrolysis is a process of thermal decomposition at moderate temperatures in the absence of oxygen. In pyrolysis, polymers are broken down into smaller products, which can be used as fuel oil or petrochemical raw material.

However, the majority of pyrolysis studies have been carried out with pure plastics. There are some unreasonable issues about their behavior in the case of plastic mixtures generated by households and industry, and others. In pyrolysis, mixed plastics are more complex than pure plastics (Vasile et al., 2001). Mixed plastics consist of various plastic types, which have different physical-chemical properties. Thus, according to the type of plastics in a plastic mixture, the quality of oil product obtained in pyrolysis is affected. In our previous work (Lee et al., 2002), we have conducted a comparison of plastic types for catalytic degradation of pure waste plastics into liquid product with spent fluid catalytic cracking (FCC) catalyst. The plastic type had an influence on the yield, molecular weight distribution and PONA distribution of liquid product obtained from catalytic degradation of pure waste plastic.

This study is aimed at studying the influence of lapse time of reaction on the characteristic of liquid product obtained from pyrolysis of waste plastic mixtures containing various plastic types at different temperatures, in a stirred semi-batch reactor. Here, the characteristics of the liquid product are described by cumulative amount distribution, molecular weight distribution and PONA distribution, as a function of lapse time of reaction.

2. Experimental

The waste plastic samples (waste high-density polyethylene (HDPE), low-density polyethylene (LDPE), polypropylene (PP) and polystyrene (PS)) were pelletized (0.32 cm (1/8 in.) (O.D.) $\times$ a few mm (L)) in a recycling company (Daeshin Co.) in Korea. They were supplied as the reactant in this experiment. Each waste plastic sample was mixed with proportional amounts of waste HDPE:LDPE:PP:PS = 33 wt%:22 wt%:33 wt%:11 wt%, based on

200 g of reactant. The proportion of plastics in the mixture was obtained from approximate data from the Korea Environment Institute.

In order to determine the thermal behavior of the waste plastic samples, thermogravimetric analysis (TGA; Dupont 950) was performed. The samples were heated from ambient to 700 °C at a heating rate of 10 °C/min under a nitrogen stream.

The degradation experiments were carried out in a stirred semi-batch reactor (volume 1.1 L) at a constant degradation temperature under atmospheric pressure (Lee et al., 2002). The experimental conditions were as follows: a reactant amount of 200 g, a nitrogen stream of 20 cm³ min, a stirrer speed of 200 rpm and heating rate of about 9 °C/min. At the beginning of the experiment, the sample was weighed and charged into the reactor. The reactor was sealed, purged with a nitrogen stream to remove the oxygen from the reactor, and then heated at a determined temperature program. During the reaction, the gas products were vented after cooling by a condenser to 7 °C. The liquid products were measured by weight, as a function of lapse time of reaction. Each component of the liquid products was quantified and qualified by gas chromatography, using FID and MS detectors. For this analysis method, a detailed explanation was provided previously (Seo et al., 2003).

3. Results and discussion

3.1. TGA analysis

The TG curves of each polymer in general waste plastics show a similar behavior in a single step, but at different temperature ranges during thermal decomposition under the same analytical conditions, as shown in Fig. 1. PS with

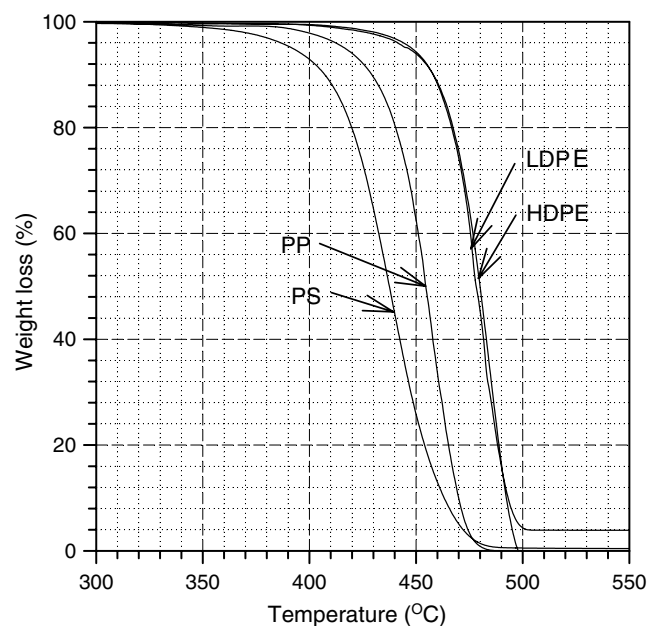


Fig. 1. TGA curves of pure waste plastics at a heating rate of 10 °C/min under nitrogen stream.

a polycyclic structure degraded within a temperature range of 350–480 °C, while PP and PE having a polyolefinic structure degraded at higher temperatures of 380–480 °C and 400–500 °C, respectively. The degradation temperatures at which the weight loss amounts to 50% (T_{50}) were about 440 °C for PS, 455 °C for PP and 480 °C for PE. Thus, the order of degradation temperature of waste thermoplastic by type of plastic was $PS < PP < HDPE < LDPE$. Among pure reactants, PS with polycyclic structure was degraded at the lowest temperature, while PP in polyolefinic polymers was degraded at a lower temperature than PE. From these results, it can be expected that plastic mixtures of different compositions will result in different product characteristics. In the pyrolysis of plastic mixture, PS can be easily degraded into light products, whereas PE will be difficult to degrade.

3.2. Pyrolysis at 350 °C

The effect of lapse time of reaction on the characteristics of liquid product obtained from the pyrolysis of waste plastic mixtures at both 350 and 400 °C was investigated. Firstly, the curves in Fig. 2 show the cumulative yield of liquid product as a function of lapse time of reaction. This result was clearly different at both temperatures. The cumulative yield of liquid product at 400 °C sharply increased with increasing lapse time of reaction, whereas the cumulative yield at 350 °C increased very slowly. This means that the degradation temperature of 400 °C in the pyrolysis of plastic mixture was good, but not as good at 350 °C.

The curves in Fig. 3 show the PONA composition of liquid product from pyrolysis of a waste plastic mixture

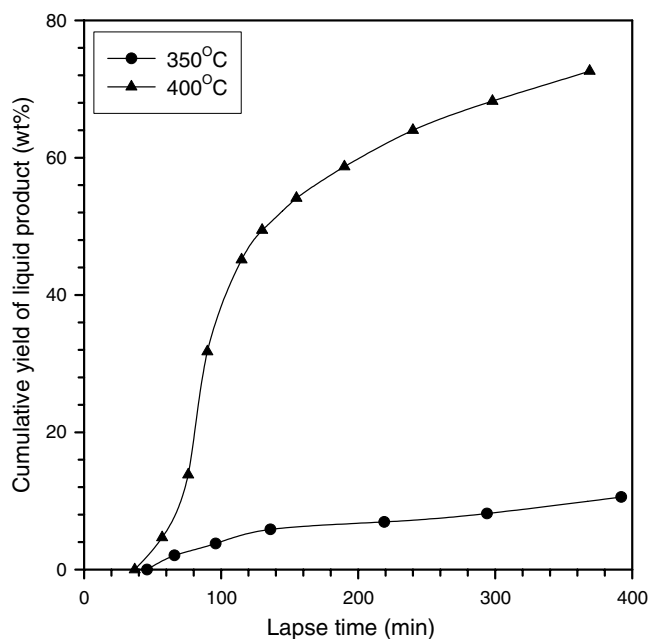


Fig. 2. Cumulative yield of liquid product as a function of lapse time of reaction for pyrolysis of a waste plastic mixture at 350 and 400 °C.

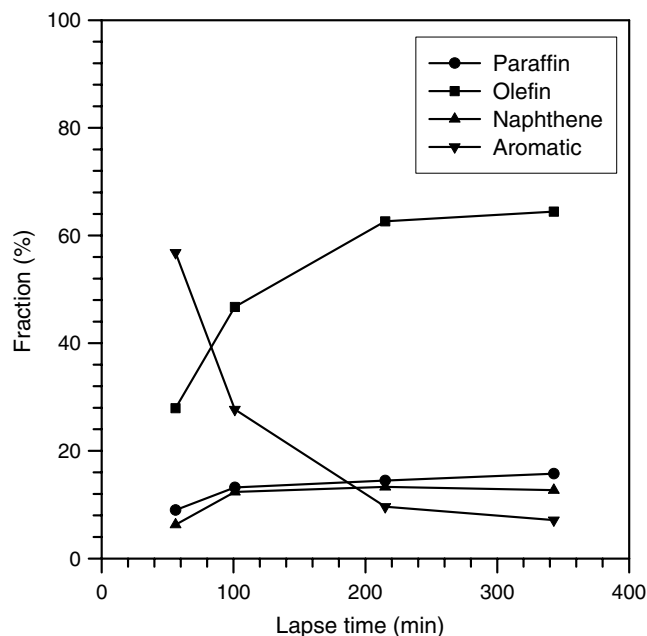


Fig. 3. Paraffin, olefin, naphthene and aromatic distribution of liquid product as a function of lapse time of reaction for pyrolysis of a waste plastic mixture at 350 °C.

as a function of lapse time of reaction. This pyrolysis process was conducted at a relatively low degradation temperature of 350 °C. The low temperature in pyrolysis will result in a much different characteristic of the oil product as compared to that produced at a high temperature. In the initial reaction (at around 50 min of lapse time), the fractions of liquid paraffin, olefin, naphthene and aromatic products were about 10%, 25%, 5% and 60%, respectively. The main product was aromatic components with about a 60% fraction. On the contrary, naphthene and straight hydrocarbon products showed low fractions. It can be explained that the initial pyrolysis using the semi-batch reactor mainly occurred by pyrolysis of the PS contained in the mixed plastics, due to the high fraction of aromatic products in the liquid PONA products. However, according to the progressive increase in the lapse time of reaction, the fraction of aromatic products decreased, whereas that of olefin products increased. The olefin products will be mainly obtained by pyrolysis of polyolefinic plastics in the plastic mixture. Among polyolefinic polymers, PP had a lower degradation temperature range than PE, as shown by the TGA curves (Fig. 1). Thus, it was demonstrated that for polyolefinic plastics at a relatively low temperature, PP was more degraded than PE. Also, in the range of lapse time of above 200 min, the fractions of paraffin, olefin, naphthene and aromatics products was almost unchanged with further increase of lapse time, consisting of about 15% paraffin, 65% olefins, 15% naphthene and 10% aromatic fractions. Here, the main product was olefin components (about 65%). This means that the pyrolysis at 350 °C was mainly determined by the degradation of PP and not PE, which has a higher degradation temperature. If PE in

the plastic mixture would have been well degraded at this temperature, the fraction of paraffin rather than olefin components would have increased (Lee and Shin, 2004).

The molecular weight distributions of liquid PONA products at different lapse times and 350 °C are shown in Fig. 4. Among them, Fig. 4(a) shows molecular weight dis-

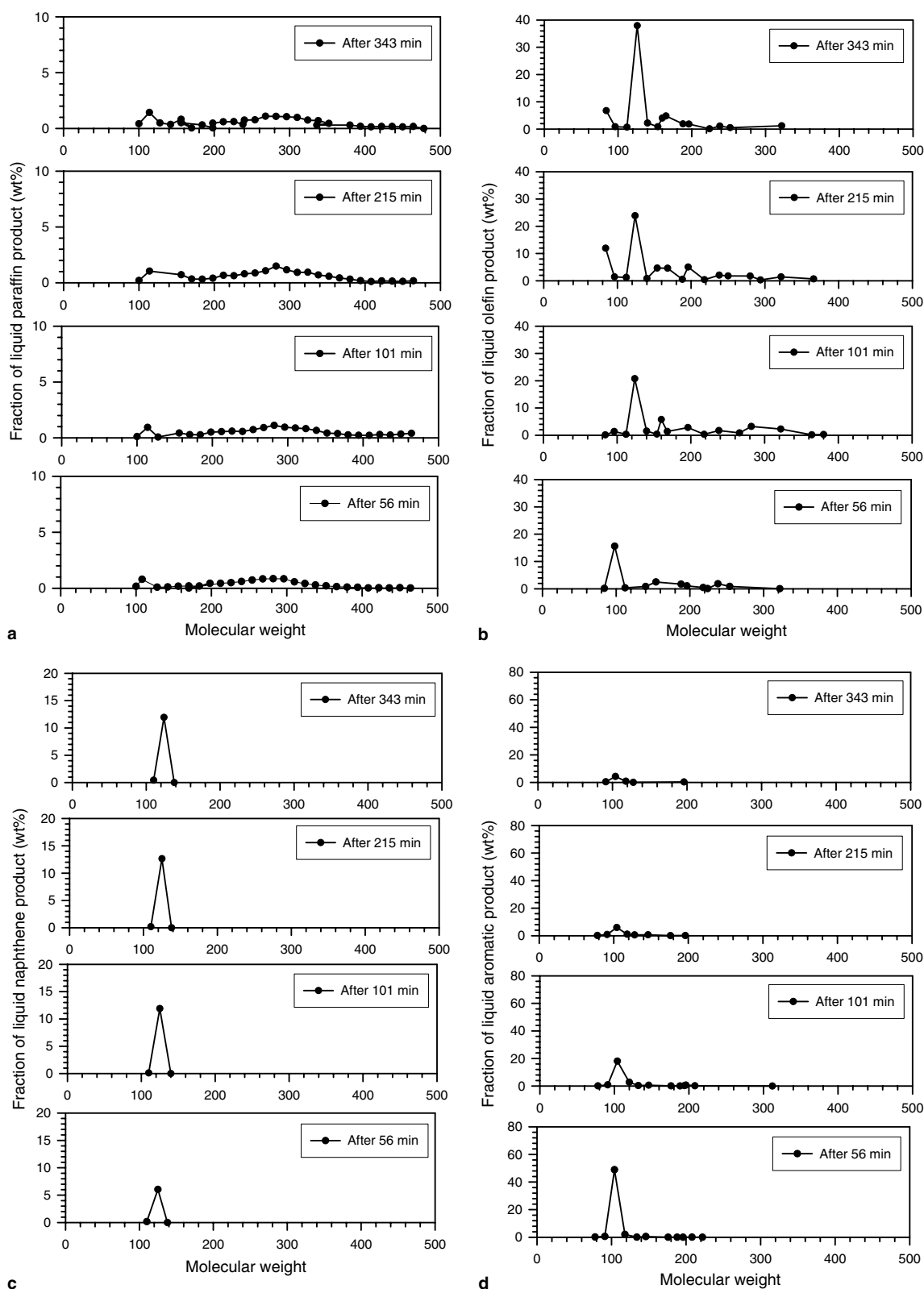


Fig. 4. Molecular weight distributions of liquid paraffin (a), olefin (b), naphthene (c) and aromatic (d) product at different lapse time for pyrolysis of a waste plastic mixture at 350 °C.

tributions of liquid paraffin products. They consisted of a wide molecular weight distribution with a low fraction in the whole range of molecular weight, due to a random scission mechanism in the degradation of polymer. With an increase of lapse time of reaction, the distribution of paraffin product fractions did not change very much. These results mean that paraffin products in the pyrolysis of plastic mixtures at this temperature were poorly produced.

On the other hand, in Fig. 4(b) liquid olefin products were produced with a high fraction and their distribution had a wide molecular weight distribution up to a molecular weight of 400, like that of paraffin products shown in Fig. 4(a), which is obtained by a random scission mechanism. The olefin products in straight hydrocarbons were produced in higher concentrations than paraffin products, occurring by hydrogenation of olefin intermediates (Miscolczy et al., 2004). Moreover, the distributions of olefin products show that light olefin components with molecular weights of around 120 were highest and the rest in molecular weight distribution was produced in very small amounts. At increasing lapse time of reaction, the fraction of these light olefin components gradually increased from about 15% to about 40%. This means that the olefin product at low temperature was enhanced by an increase of lapse time in the reactor. From these results, we can find that the degradation mechanism in the pyrolysis of plastic mixtures consisting of polyolefinic polymer types also was mainly an end-chain scission mechanism.

Molecular weight distributions of liquid naphthene products at different lapse times of reaction are presented in Fig. 4(c). The main naphthene product only appeared at a molecular weight of around 120 as light liquid product with one cyclic structure, which will be formed by cyclization of light straight hydrocarbons produced in the pyrolysis of polyolefinic polymer. This can be explained by the peaks in the light olefin products with a molecular weight of around 120, as shown in Fig. 4(b). As the lapse time of reaction was increased, the fraction of light naphthene and olefin products with a molecular weight of around 120 gradually increased. That is, the production trend of both naphthene and olefin components with a similar molecular weight was very similar. Accordingly, the degradation mechanism in the pyrolysis of the mixture simultaneously resulted in olefin products by primary cracking of polymer and naphthene products by cyclization of olefin intermediates. Other naphthene components except around 120 of molecular weight, as shown in Fig. 4(c), were almost absent.

The curves in Fig. 4(d) show the molecular weight distributions of liquid aromatic products. The main products in these distributions were molecular weights of around 100 as C₈ components, which had one benzene ring structure such as styrene and ethylbenzene, etc. As products with cyclic structures, this distribution clearly differed from that of naphthene product of similar molecular weight and at equal lapse time of reaction, as shown in Fig. 4(c). From this result, we can show that these two types of cyclic prod-

ucts are produced by a different degradation mechanism in the pyrolysis of polymer mixtures. In initial pyrolysis of the mixture, the C₈ aromatic products showed a very high fraction with about 50% in liquid products, which is mostly obtained from pyrolysis of PS in the mixture (Lee and Shin, 2004). At increasing lapse time of reaction, the fraction of aromatic products sharply decreased to below 10%. It was demonstrated that PS in the plastic mixture was pyrolyzed very fast into light liquid products. With a function of lapse time of reaction, this trend showed a reverse relationship with those of olefin and naphthene products. Therefore, the characteristics of the oil product that is obtained by pyrolysis of a plastic mixture are influenced by the type of plastic in the mixture.

3.3. Pyrolysis at 400 °C

In order to compare the characteristics of the oil product at both low and high degradation temperatures, pyrolysis was conducted at 400 °C as a high degradation temperature. At this temperature, the plastic mixture would be converted well into a liquid product. The curves in Fig. 5 show the distribution of the liquid PONA product for the pyrolysis of a waste plastic mixture at 400 °C, as a function of lapse time of reaction. In the initial lapse time of reaction, the fractions of liquid paraffin, olefin, naphthene and aromatic products were about 5%, 15%, 0% and 80%, respectively. The main liquid product consisted of aromatic components, whereas naphthene and paraffin components were produced in very small amounts. This trend was very similar to that of the PONA distribution at 350 °C, as shown in Fig. 3. Aromatic products were mostly obtained from the pyrolysis of PS in the mixture.

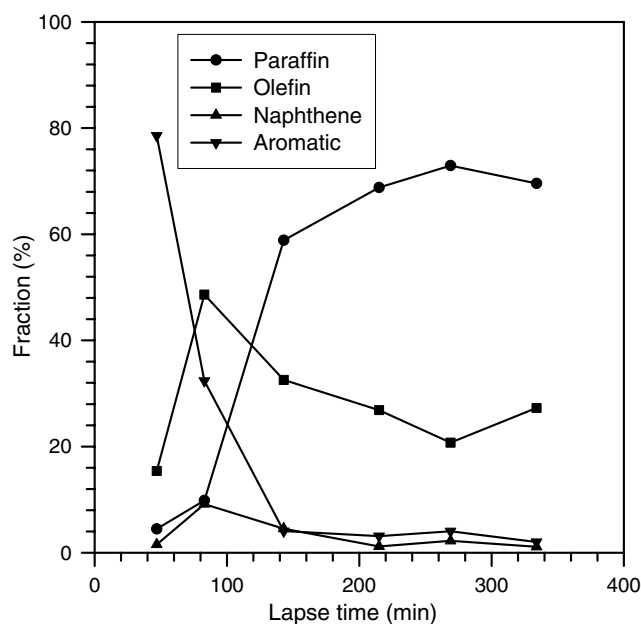


Fig. 5. Paraffin, olefin, naphthene and aromatic distribution of liquid product as a function of lapse time of reaction for pyrolysis of a waste plastic mixture at 400 °C.

As the lapse time of reaction was increased from about 50 min to about 80 min, the fraction of aromatic product sharply decreased from about 80% to about 30%, whereas that of olefin products increased from about 15% to about 50%. This phenomenon indicates that the degradation of polyolefinic plastic, rather than that of polycyclic plastic, in the mixture enhanced the olefin products by means of primary degradation of polymer. A similar trend appeared during the pyrolysis of the mixture at 350 °C, as shown in Fig. 3. On the other hand, above 80 min of lapse time the PONA distribution of liquid product changed to a large extent with lapse time. With the increase of lapse time, from about 80 min to about 140 min, the fraction of paraffin products was sharply increased from about 15% to about 60%, whereas those of olefin and aromatic components decreased from about 50% to about 35%, and from about 35% to below 10%, respectively. Moreover, after

140 min of lapse time, this distribution was not changed with further increase of lapse time, with the main product consisting of paraffin components. Here, the increase of paraffin products was accompanied by a decrease of olefin products, while this is not the case at 350 °C (Fig. 3). This result can be explained by different mechanisms at different degradation temperatures. It is supposed that some plastic material is not sufficiently degraded at 350 °C. From TGA analysis as shown in Fig. 1, PE is the most difficult material to degrade in the mixture. It can be deduced that pyrolysis of PE contained in the mixture degrades the material well into liquid product at 400 °C and not at 350 °C. Thus, a temperature of 400 °C in the pyrolysis of a plastic mixture is a better condition to obtain liquid product.

For the pyrolysis of a waste plastic mixture at 400 °C, molecular weight distributions of liquid paraffin products at different lapse times are shown in Fig. 6(a). At the initial

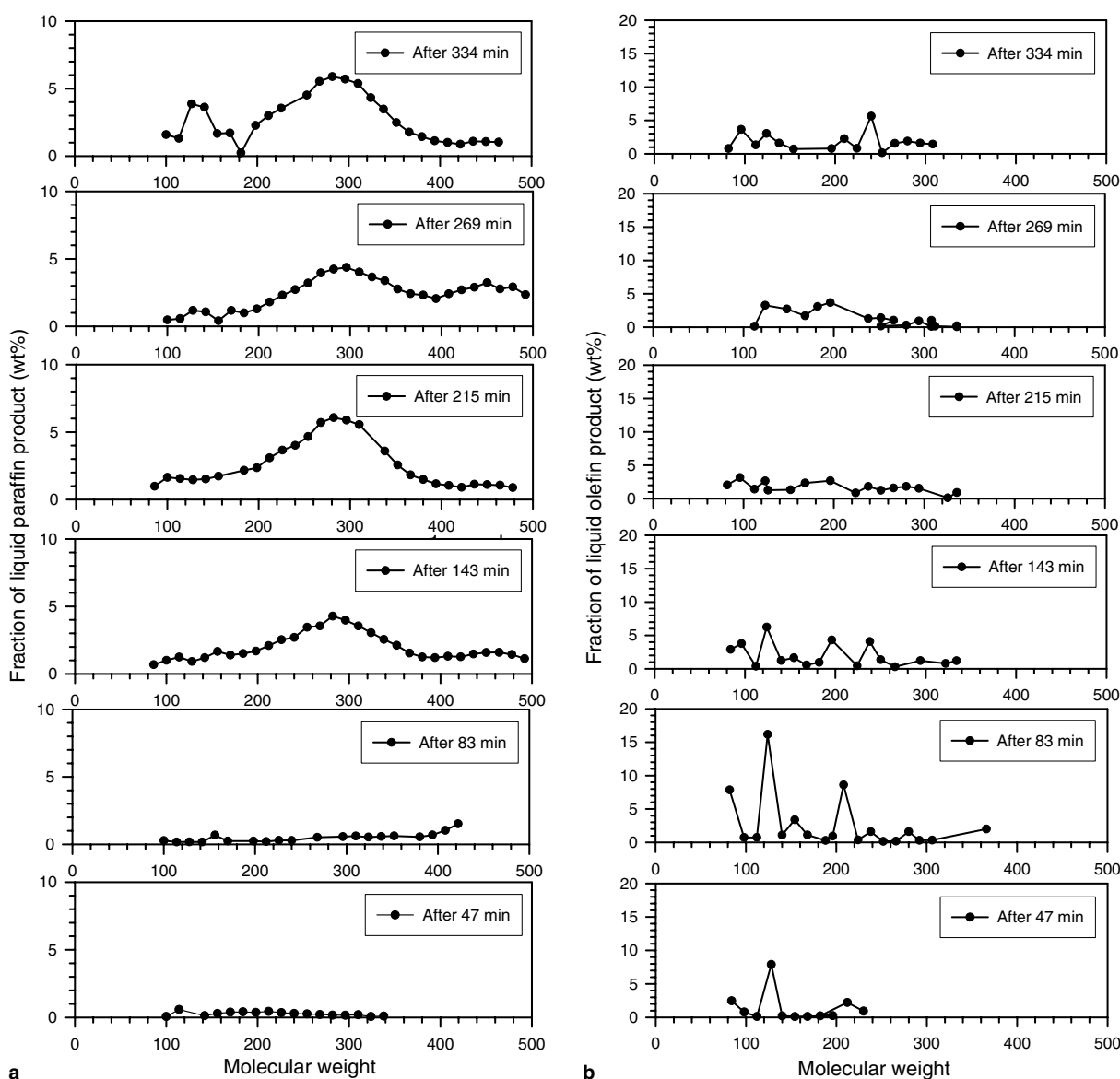


Fig. 6. Molecular weight distributions of liquid paraffin (a), olefin (b), naphthene (c) and aromatic (d) products at different lapse time for pyrolysis of a waste plastic mixture at 400 °C.

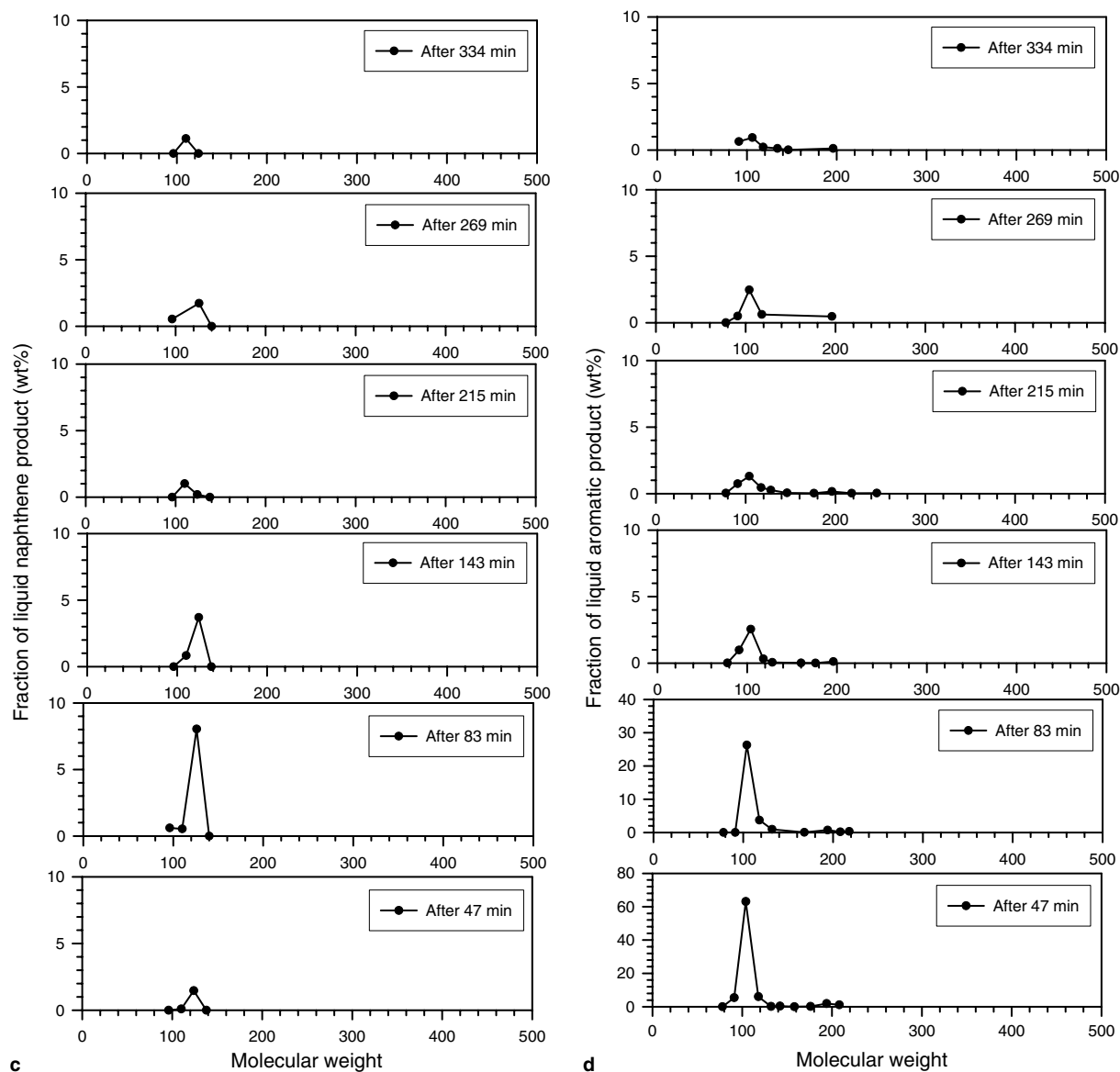


Fig 6. (continued)

lapse time, liquid paraffin products were produced in small amounts with a wide molecular weight distribution. As the lapse time of the reaction increased from 83 to 143 min, liquid paraffin products were produced in higher concentrations. Moreover, after 215 min of lapse time of reaction, liquid paraffin products were sufficiently obtained. With an increase of the lapse time of reaction, the production of liquid paraffin products was progressively enhanced. This phenomenon did not appear at 350 °C, as shown in Fig. 4(a). Pyrolysis at 350 °C showed the paraffin product to be present in very small amounts. As the degradation temperature was high and the lapse time of reaction also was increased, paraffin products were more enhanced. It can be deduced that paraffin products were mainly obtained by pyrolysis of the PE in the mixture, because PE material is the most difficult material to pyrolyze among the plastic types in the mixture that were used in

this experiment. These distributions had a wide molecular weight distribution up to a molecular weight of 500 and a large quantity between molecular weights of 200 and 400, which suggests that from this distribution the degradation proceeds mainly by a random scission mechanism. Also, the main product for pyrolysis of the mixture at 400 °C was paraffin components in liquid PONA products.

The curves in Fig. 6(b) shows molecular weight distributions of liquid olefin products at different lapse times. This shows a wide distribution of molecular weight up to 400, like that of paraffin products, as shown in Fig. 6(a), which were mainly obtained by a random scission mechanism. However, molecular weight distributions in paraffin and olefin products that were mainly produced from pyrolysis of polyolefinic polymer were very different as a function of lapse time of reaction. In the case of paraffin products (Fig. 6(a)), their quantity was gradually increased with

the increase of lapse time, whereas the olefin products showed a maximum quantity with mainly light liquid components at 83 min of lapse time as the initial reaction time, due to end-chain scission of large straight hydrocarbon intermediates. This result indicates that the paraffin and olefin products in the pyrolysis of the waste plastic mixture were produced by a different degradation mechanism, which first resulted in the formation of olefin products and then paraffin products.

As a function of lapse time of reaction, molecular weight distributions of olefin products at 400 °C showed a different result compared to those at 350 °C, as shown in Fig. 4(b). The case at 400 °C produced more light olefin products at 83 min as a initial lapse time, whereas those at 350 °C were gradually increased with increase of lapse time. It can be explained that a high degradation temperature results in the very rapid formation of olefin products by means of a primary cracking reaction of polymer.

As a cyclic product, Fig. 6(c) shows molecular weight distributions of liquid naphthene products at different lapse times. Naphthene products showed a very narrow distribution with a molecular weight of around 120 in the whole range of lapse time of reaction, consisting of light liquid components with one cyclic structure. These distributions were very different from those of paraffin and olefin products described previously. That is, naphthene products consisted of simple components compared to those of paraffin and olefin products. With increasing lapse time of reaction, the fraction of naphthene products was obtained with a little below 2% fraction at initial lapse time, a maximum amount at 83 min of lapse time and was gradually decreased after 83 min of lapse time, which is similar to that of olefin product as shown in Fig. 6(b). It can be explained that the pyrolysis of a waste plastic mixture simultaneously produced light olefin and naphthene products. It is suggested that large paraffin and olefin intermediates from the pyrolysis of a waste plastic mixture are converted into light olefin products by a β -scission mechanism and also light naphthene products by cyclization of cracked light straight hydrocarbons.

Also, molecular weight distributions of aromatic products are shown in Fig. 6(d). The aromatic products showed a narrow molecular weight distribution with light liquid products, which is similar to that of naphthene products as shown in Fig. 6(c). These cyclic products showed a simple structure product with a molecular weight of around 100 for pyrolysis of a mixture that consists of different polymer structures. However, the distribution in two cases clearly differed as a function of lapse time of reaction. Naphthene products showed a maximum fraction of about 8% at a lapse time of 83 min, whereas aromatic products had a maximum fraction of above 60% at the initial lapse time. Moreover, aromatic products were produced in higher concentrations than naphthene products. These results mean that these two cyclic product types showed a different degradation mechanism during pyrolysis of a

waste plastic mixture. Aromatic products were mostly light liquid products with one benzene ring of C₈ components by means of end-chain scission of hydrocarbons, and sharply decreased with an increase of lapse time of reaction, after the initial lapse time of reaction had its maximum fraction. Also, this fraction at 400 °C was higher than that at 350 °C, as shown in Fig. 4(d). From these results, it can be explained that because the aromatic products in PONA components showed the highest fraction at the initial lapse time of reaction, PS in the mixture was first degraded into a liquid product.

4. Conclusions

The characteristics of the liquid product obtained from the pyrolysis of a waste plastic mixture (HDPE:LDPE:PP:PS = 3:2:3:1) at low (350 °C) and high (400 °C) degradation temperatures were investigated as a function of lapse time of reaction, in a stirred semi-batch reactor. PONA distributions of liquid product and molecular weight distribution in PONA components changed with lapse time of reaction and also were different at low and high degradation temperatures.

The cumulative yield of liquid product at 400 °C sharply increased with increasing lapse time of reaction, compared to that at 350 °C. Thus, the degradation temperature of 400 °C in the pyrolysis of plastic mixture was good, but not as good at 350 °C. For liquid PONA distributions, in the initial range of lapse times the main product was aromatic product, regardless of degradation temperature of 350 or 400 °C. As the lapse time of reaction increased, the aromatic product fraction decreased, whereas the olefin product fraction increased, due to more primary degradation of the polyolefinic polymer in the mixture. However, with further increase of lapse time, the main products at a high temperature consisted of paraffin components with a fraction above 70% as liquid products, compared to olefin components mainly obtained at low temperature.

From the molecular weight distribution of liquid PONA components, the experiments demonstrated that the paraffin and olefin products had a wide distribution by mainly random scission of polymer, but in the case of olefin products were produced by an end-chain scission mechanism as well as random scission mechanism, as evidenced by much more light olefin products. With respect to the lapse time of reaction, the fraction of light olefin products with molecular weights of around 120 as a main product had a similar trend with that of light naphthene products, in which the degradation temperatures of 350 and 400 °C showed their maximum fraction at a lapse time of 353 and 83 min, respectively. This means that light naphthene products in cyclic products were produced by cyclization of light olefin intermediates. However, aromatic products were mainly obtained by end-chain scission of PS contained in mixture, which were light aromatic products with a molecular weight of around 100. These naphthene and aromatic

products showed a narrow molecular weight distribution, but obtained by a different reaction mechanism in the pyrolysis of a mixture.

In conclusion, the lapse time of reaction at different degradation temperatures in a tank reactor to obtain the alternative fuel oil through the pyrolysis of a waste plastic mixture had a considerable effect on the qualitative properties of the liquid products.

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